19 Batch reactor energy balance

Let's derive the energy balance for the batch reactor. Starting from the general energy balance equation:

$$\frac{d\hat{E}_{sys}}{dt} = \dot{Q} + \dot{W}_s + \sum F_i \cancel{H_i}\big|_{in} - \sum F_i \cancel{H_i}\big|_{out}$$

$$\quad 0 \qquad\qquad 0$$

Where $\hat{E}_{sys}$ is the total energy of all components in the system

$$\hat{E}_{sys} = \sum N_i \epsilon_i = \sum N_i U_i + \sum N_i KE_i + \sum N_i PE_i$$

internal kinetic potential

Simplifying, $\hat{E}_{sys} = \sum N_i U_i$ since we ultimately seek $\Delta \hat{E}_{sys}$ & $\Delta U \gg \Delta KE$ & ΔPE. $U_i = H_i - PV$, Since we ultimately seek ΔE_{sys}, ΔPV is also $\ll \Delta U$ & hence PV can be neglected.

So, $\hat{E}_{sys} = \sum N_i H_i$

& $\frac{d\hat{E}_{sys}}{dt} = \frac{d}{dt}\sum N_i H_i$

Applying the product rule $\frac{d\hat{E}_{sys}}{dt} = \left[\sum N_i \frac{dH_i}{dt} + \sum H_i \frac{dN_i}{dt}\right]_{sys}$

Recalling that $H_i = H_i^o(T_{ref}) + \int_{T_{ref}}^{T} C_{P_i} dT$

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$$\frac{dH_i}{dt} = \frac{d}{dt}\left(\cancel{H_i^o(T_{ref})} + \int_{T_{ref}}^{T} C_{P_i}\, dT\right) = \frac{d}{dt}\int_{T_{ref}}^{T} C_{P_i}\, dT = C_{P_i}\frac{dT}{dt}$$

$$\qquad\quad 0$$

(assuming C_{P_i} are constant)

Then: $\dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{P_i} \frac{dT}{dt} + \sum H_i \frac{dN_i}{dt}$

Recall $\frac{dN_i}{dt}$ is the mole balance and equals $r_i V$

Then $\dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{P_i} \frac{dT}{dt} + \sum H_i r_i V.$ $\quad r_i = \frac{\nu_i}{-\nu_A} r_A = \frac{\nu_i}{|\nu_A|}(-r_A)$

$$\sum N_i C_{P_i} \frac{dT}{dt} + \sum H_i \frac{\nu_i}{|\nu_A|}(-r_A V)$$

Recalling that $\sum \nu_i H_i = \Delta H^{rxn}$ per mole A the energy balance becomes

$$\dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{pi} \frac{dT}{dt} + \Delta H^{rxn}(-r_A V)$$

Solving for $\frac{dT}{dt}$:

$$\frac{dT}{dT} = \frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{\sum N_i C_{pi}}$$

Where $(\Delta H^{rxn})(r_A V)$ is heat generated $= \dot{Q}_{gb}$

And $\dot{Q}$ is heat removed $= \dot{Q}_{rb}$

$$\dot{Q} = UA(T_a - T)$$

Batch reactor energy balance written in terms of X:

Recall $N_i = N_{A0}\left(\Theta_i - \frac{\nu_i}{\nu_A} X\right)$

Then, $\frac{dT}{dT} = $

$$\frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{N_{A0} \sum C_{p_i}\left(\Theta_i - \frac{\nu_i}{\nu_A} X\right)}$$

Recall $\sum \frac{\nu_i}{|\nu_A|} C_{P_i} = \Delta C_p$ per mole A

$$\boxed{\text{Then, } \frac{dT}{dT} = \frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{N_{A0}\left(\sum \Theta_i C_{p_i} + \Delta C_p X\right)}}$$

where often we neglect $\dot{W}_s$

Let's do some examples:

- The reaction A + B --> C is carried out in a batch reactor. C_{PA} = 35 cal/mol/K, C_{PB} = 18 cal/mol/K, C_{PC} = 46 cal/mol/K. Determine ΔC_P.

$$\Delta C_p = \sum \frac{\nu_i}{|\nu_A|} C_{pi} = \frac{-1}{1} C_{pA} + \frac{-1}{1} C_{pB} + \frac{1}{1} C_{pC} =$$

$$= C_{pC} - C_{pA} - C_{pB} = 46 \frac{cal}{mol\,K} - 35 \frac{cal}{mol\,K} - 18 \frac{cal}{mol\,K}$$

$$= -7 \frac{cal}{mol\,K}$$

- For the same reaction, the inlet molar flowrates are 1 mol A/min, 2 mol B/min, and 0 mol C/min. Determine Θ_i.

$$\Theta_i = \frac{F_{i0}}{F_{A0}}$$ assuming A is the LR

$$\Theta_A = 1, \quad \Theta_B = 2, \quad \Theta_C = 0$$

- The reaction is carried out adiabatically. What is the energy balance?

Two options

① $$\frac{dT}{dt} = \frac{(\Delta H_{rxn})(r_A V)}{N_{A0}\left(\sum \Theta_i C_{pi} + \Delta C_p X\right)}$$

② $$T = T_0 + \frac{(-\Delta H_{rxn})X}{\sum \Theta_i C_{pi} + \Delta C_p X}$$